

SAFETY IN DESIGN & OPERATION - OIL REFINERY

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RISKS AND HAZARDS



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HAZARDS AND RISKS

HAZARD

- *Hazard is a potential source of harm or adverse effect on a system.*

RISK

- *Risk is the likelihood that the system may be harmed if exposed to a hazard.*

POTENTIAL REFINERY HAZARDS

- Hydrocarbon (Gases and Liquids)
 - Steam
 - Compressed air
 - Water
- Electricity and electrical equipments
 - High temperature & pressure

MIS-HANDLING OF ANY THE ABOVE CAN LEAD TO ACCIDENT.



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SOME MAJOR DISASTERS



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MAJOR DISASTERS

BP TEXAS REFINERY



- BP AMOCO Refinery is on 1,200 acres with 30 refinery units and is 71 years old.
- 1800 people work at the refinery plus contractors
- It is BP's largest plant, and the USA's third largest refinery, processing 460,000 barrels of crude oil/day, around 3% of US Gasoline supplies



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MAJOR DISASTERS

BP Texas Refinery – The Aftermath



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MAJOR DISASTERS

Bruncefield, UK



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MAJOR DISASTERS

Piper Alpha



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MAJOR DISASTERS

IT CAN HAPPEN TO US!!



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RISK MANAGEMENT



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WHY RISK MANAGEMENT

- Risk Management is the process of identifying, analyzing and responding to risk factors throughout the life of a project and in the best interests of its objectives. Proper risk management implies control of possible future events and is proactive rather than reactive.
- Necessity of risk management is to identify, reduce and plan for the foreseen risks attached with the project.
- Risk Management systems not just identify risk but quantify it and predict its impact.



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RISK RESPONSE

Risk Response generally includes:

AVOIDANCE

- Eliminating a specific threat, usually by eliminating the cause.

MITIGATION

- Reducing the expected monetary value of a risk event by reducing the probability of occurrence.

ACCEPTANCE

- Accepting the consequences of the risk. by developing a contingency plan to execute should the risk event occur.



CONTROL MEASURES

Control measures include actions that can be taken to reduce the potential of exposure to the hazard, or the control measure could be to remove the hazard or to reduce the likelihood of the risk of the exposure to that hazard being realised.

Elimination of hazard

Substitution of hazard with lesser risks

Isolation of the hazard

Use of engineering controls

Use of administrative tools

Use of Personal Protection Equipment



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Hazard Controls



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HAZARD CONTROL

PREVENTION CONTROLS

Elimination

Substitution

Alarms & Procedures

Engineering

Passive Devices

Prevention of “Escalation from other incident”

MITIGATION CONTROLS

Ignition Prevention

Alarms and Procedures

Mitigation

Emergency Response

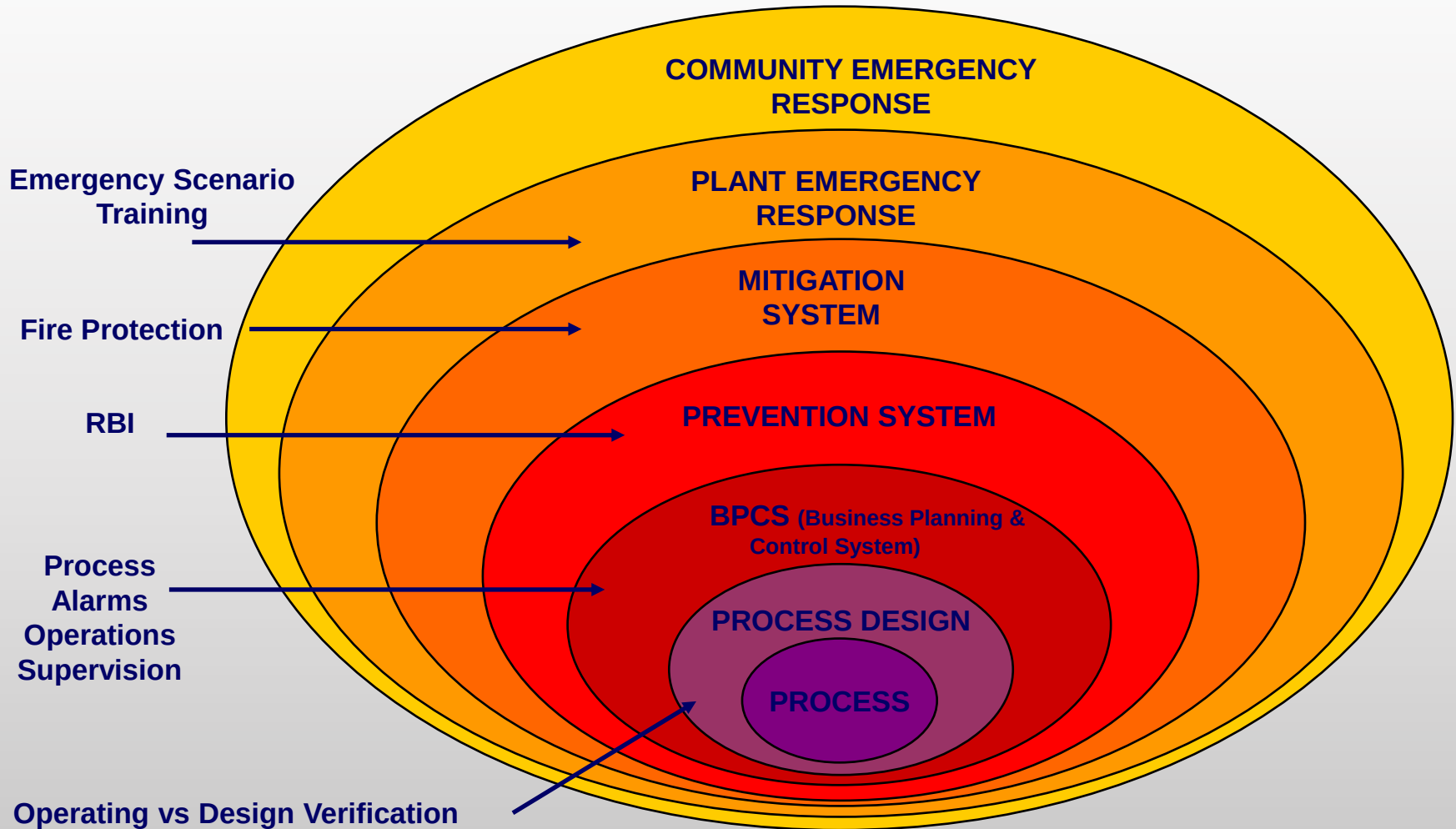
Prevention of “Escalation to other vessel”



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HAZARD CONTROL



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SAFETY AT VARIOUS STAGES



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ROOT CAUSES OF INCIDENTS

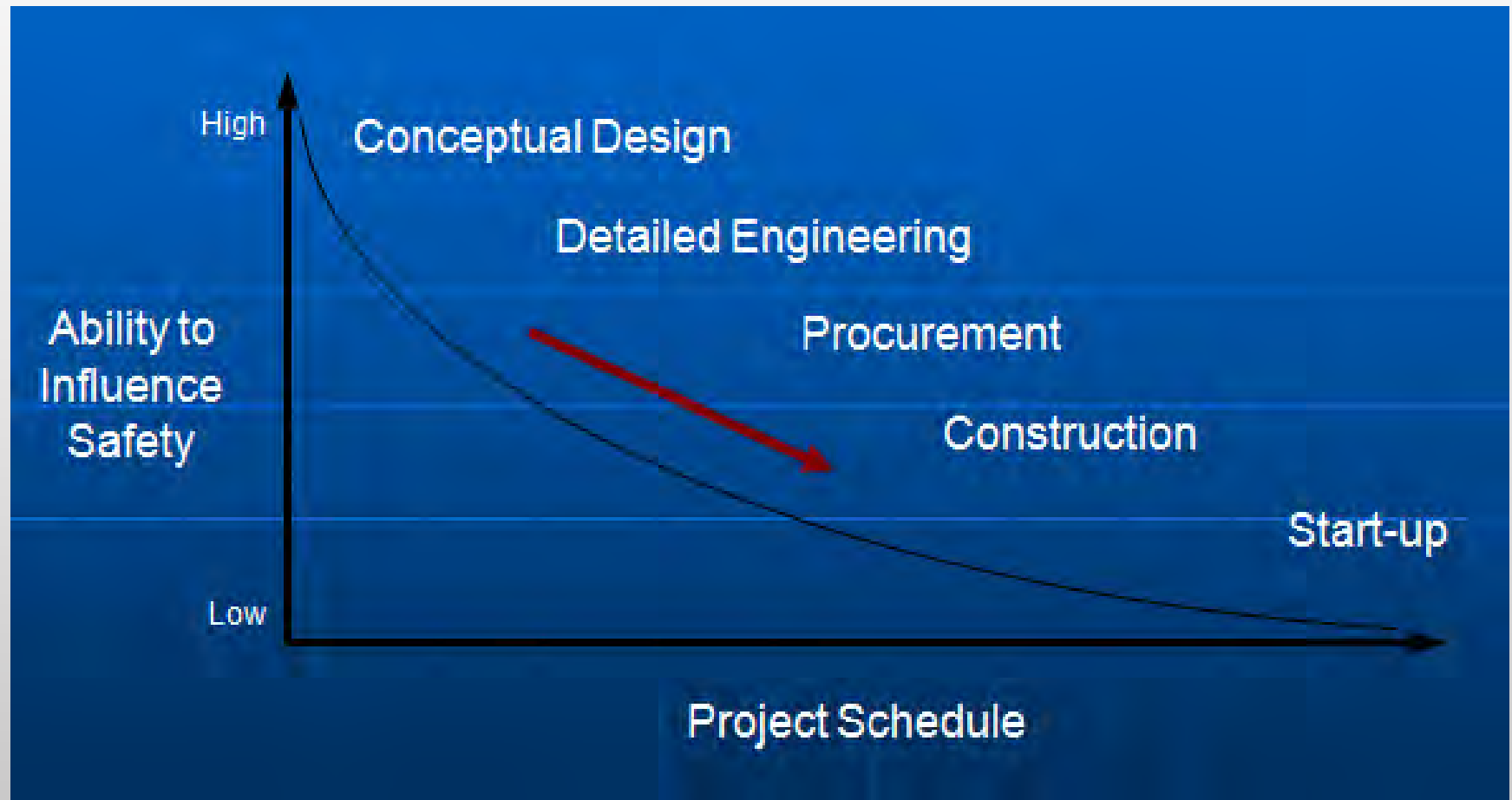
- Inadequate construction planning
- Lack of proper training
- Deficient enforcement of training
- Unsafe equipment
- Unsafe methods or sequencing
- Unsafe site conditions
- Not using safety equipment that was provided



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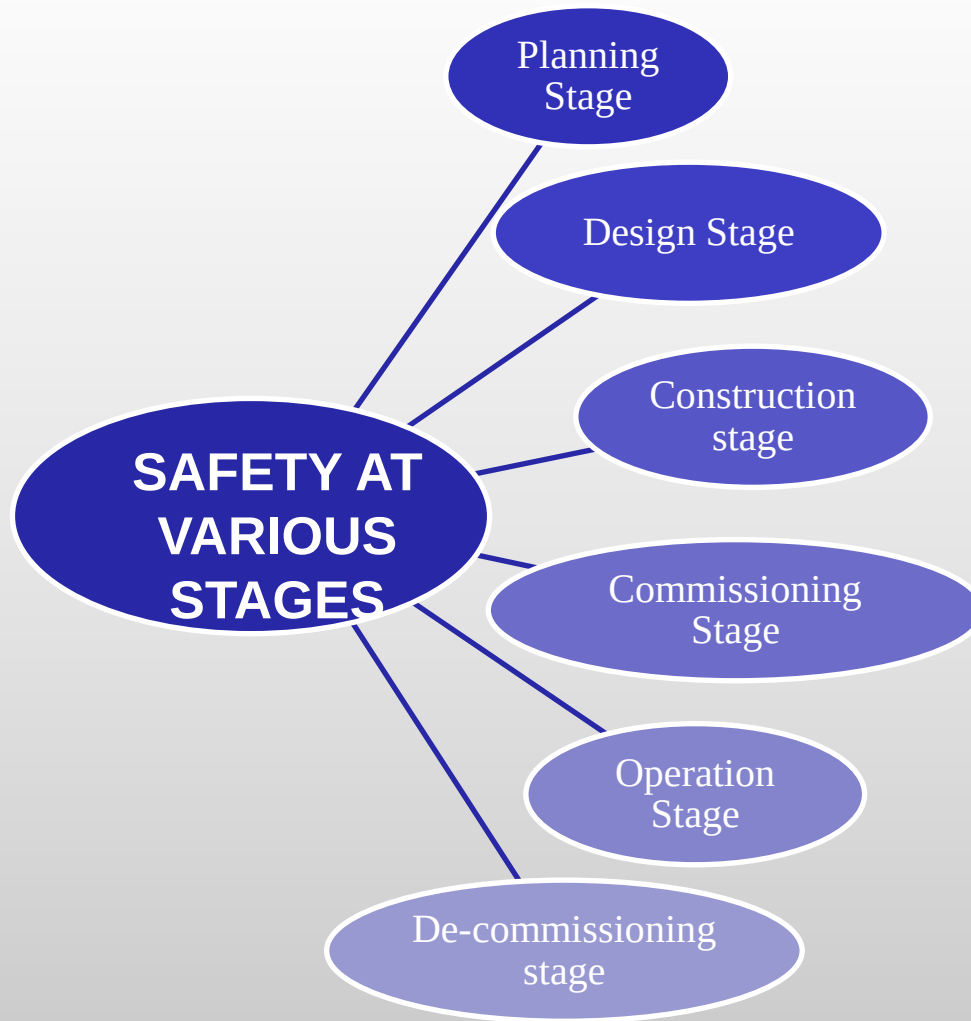
SAFETY CONSIDERATIONS



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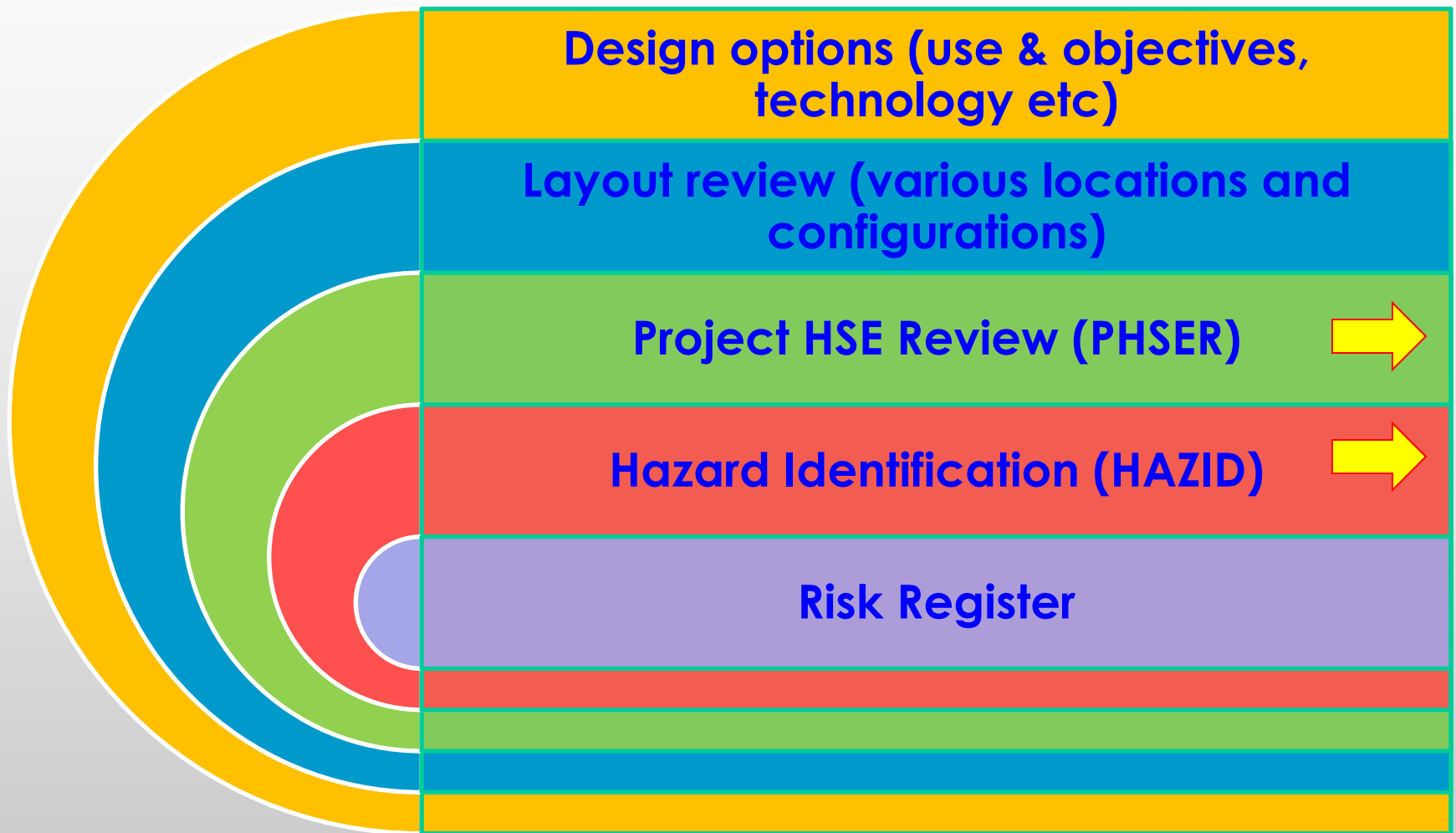
SAFETY AT VARIOUS STAGES



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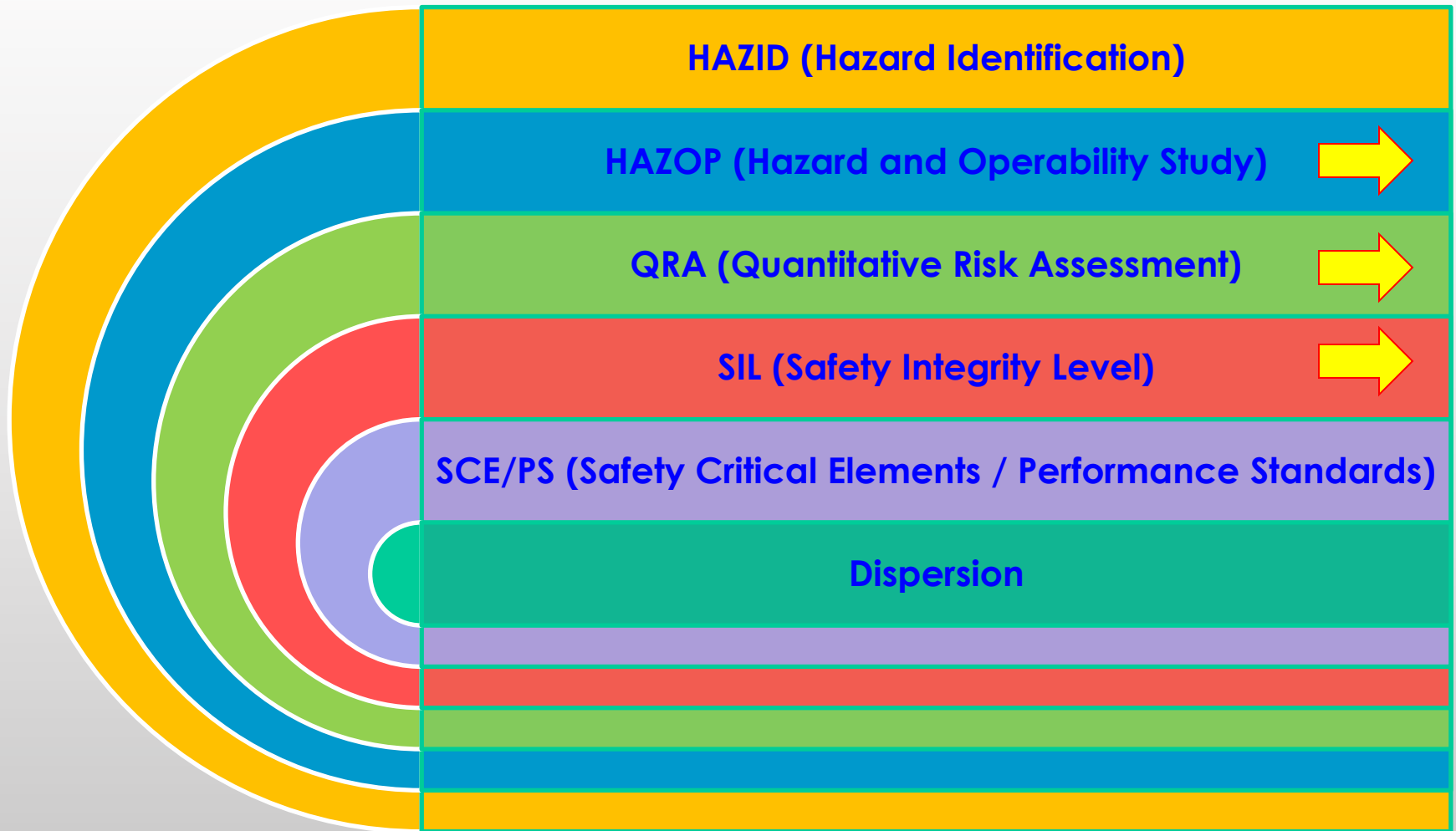
SAFETY AT PLANNING STAGES



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SAFETY AT DESIGN STAGES



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SAFETY AT DESIGN STAGES



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SAFETY AT DESIGN STAGES

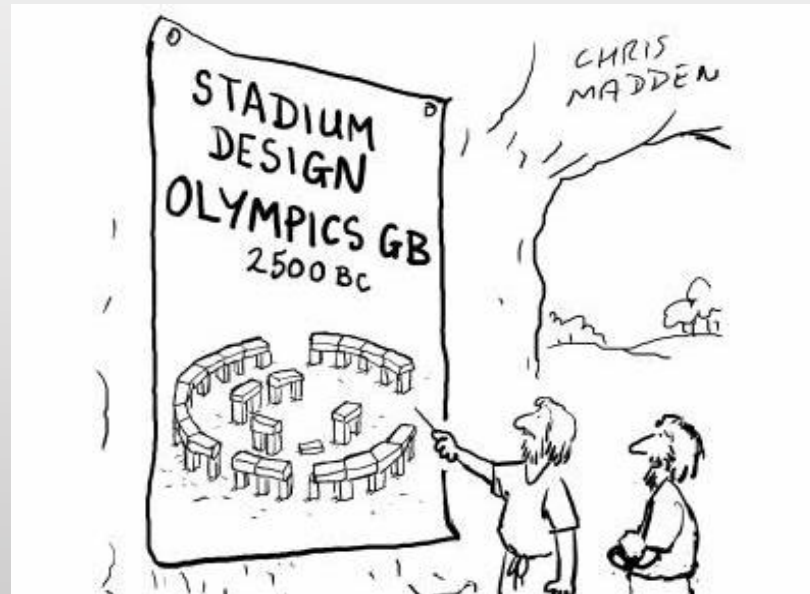
- Project HSE Review (PHSER)
- ALARP (AS low as reasonably practicable)
- RAM (Reliability, availability & maintainability analysis)
- Safety Case / COMAH (Control of major accident hazards)
- Fire Water network
- Fire Fighting System and extinguishers
- Emergency Response Plan
- Risk Register



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SAFETY CONSIDERATIONS DURING DESIGN/ ENGINEERING STAGE



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DESIGN CONSIDERATIONS

Basic design considerations:

- Design data for reliable operation of Unit
- Design, metallurgy and corrosion allowance as per intended service
- Proper selection of mechanical design considerations
- Design for minimal corrosion effects
- Proper design of tall structures
- Consideration of environmental aspects
- Plant layout with all safety considerations



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Safety considerations in design : examples

- Push buttons for stopping critical motors shall be provided at a safe location, away from the fire-zone of the respective motors.
- Gas detectors / hydro carbon detectors etc shall be provided in the critical process and offsite areas
- Flare line isolation valve at unit battery limits should be installed only in the horizontal line with stem in horizontal position
- Provision of ROVs & MOVs



Safety considerations in design : examples

- Maximizing pre-fabrication jobs
- Ensuring escape routes, adequate fire fighting facilities, safety showers etc
- Minimizing the requirement of temporary access
- Ensuring ease of operation
- Erection plan for all erection activities
- Safety interlocks
- HAZOP study



Safety considerations in design : examples

- Provision of Reinforcement pads at pipe supports at environmentally corrosive locations.
- Proper paint system suitable for the corrosivity of the area.
- Adequate pressure / surge protection, installation of TSVs & PSVs.
- All gas sampling incl. LPG sampling points should be 'closed-sampling' type.
- dP indication should be provided across demister pads of critical vessels/columns.



Safety considerations in design : examples

- Back washing facility for coolers/ condensers.
- Use of EFSW pipes in place of ERW pipes.
- Equipment and piping design with scope for inspection and maintenance
- Ensuring proper quality of fabrication jobs.
- Providing moisture barrier to mitigate corrosion under insulation.
- Piping layout, clearance from ground, hard surfacing etc.
- Protection of under ground lines



Safety considerations in design : examples

- Welding of small bore piping and tracer lines, size $\frac{3}{4}$ " and below by TIG process
- Minimum thickness of pipe for sizes up to $1\frac{1}{2}$ " should be Sch.80
- Steam drain points should be routed to a drain header and taken out of the unit area
- Fire fighting points are to be provided at higher elevation in case of tall columns, structures.
- All the reinforcement pad telltale holes should be drilled and tapped properly.



Safety considerations in design : examples

- Wherever two phase flow in piping is expected, piping design including its support system should be checked w.r.t. most adverse conditions.
- The supports to be checked for their adequacy to bear the required loads and movements of the system.
- All SS piping should have chloride free insulation or preferably should have SS foil wrapped between pipe and insulation.
- Branch connections for fire hydrants along the roads should be totally above grounds.



Safety considerations in design : examples

- Firewater and cooling water lines emerging from underground should be wrapped coated beyond the ground level upto a length of 500mm.
- Hard surfacing with a proper slope towards open drain system is to be provided beneath the offsite pipe rack area with a clear space of 500mm from bottom of the pipe.
- Interspacing between the offsite piping on the support pedestal should be such that the lines should not touch each other even after insulation (at least 3” gap after insulation).

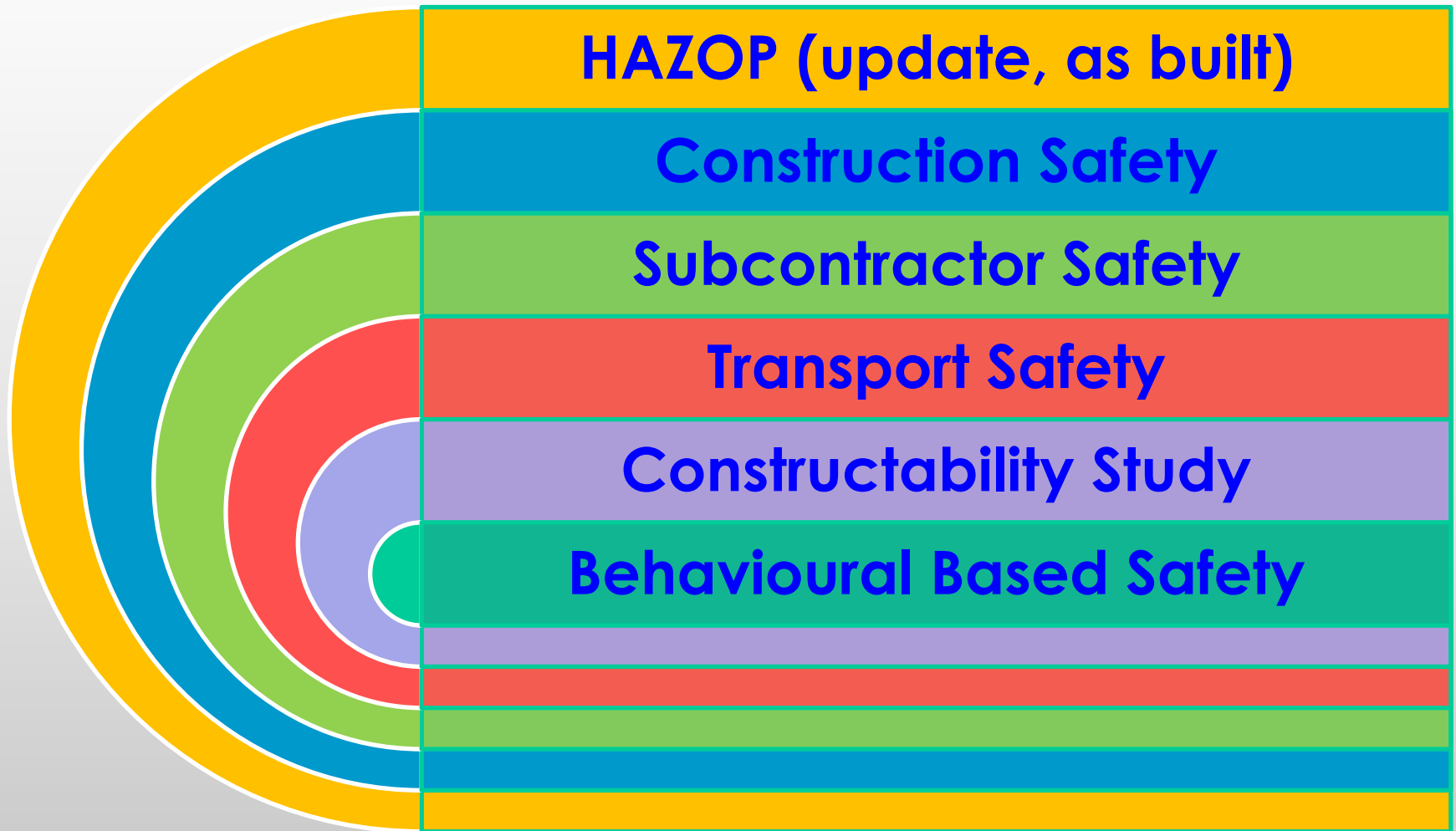


Safety considerations in design : examples

- Long lengths of vent and drain piping should be properly supported w.r.t. main pipe.
- Piping insulation ends should be properly sealed to avoid water ingress.
- Hard surfacing under the piping bay in offsite areas should be done with proper slope & drainage facility.
- Proper slope and gap should be provided in piping culverts to avoid water logging.
- In fire water lines, the hydrant tapping should not be taken from the bottom side of pipe.



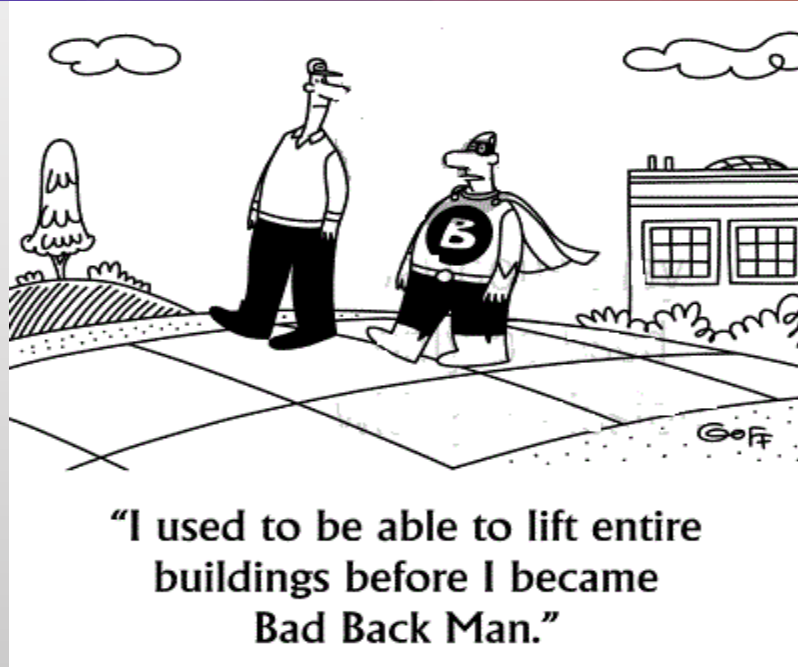
SAFETY AT CONSTRUCTION STAGES



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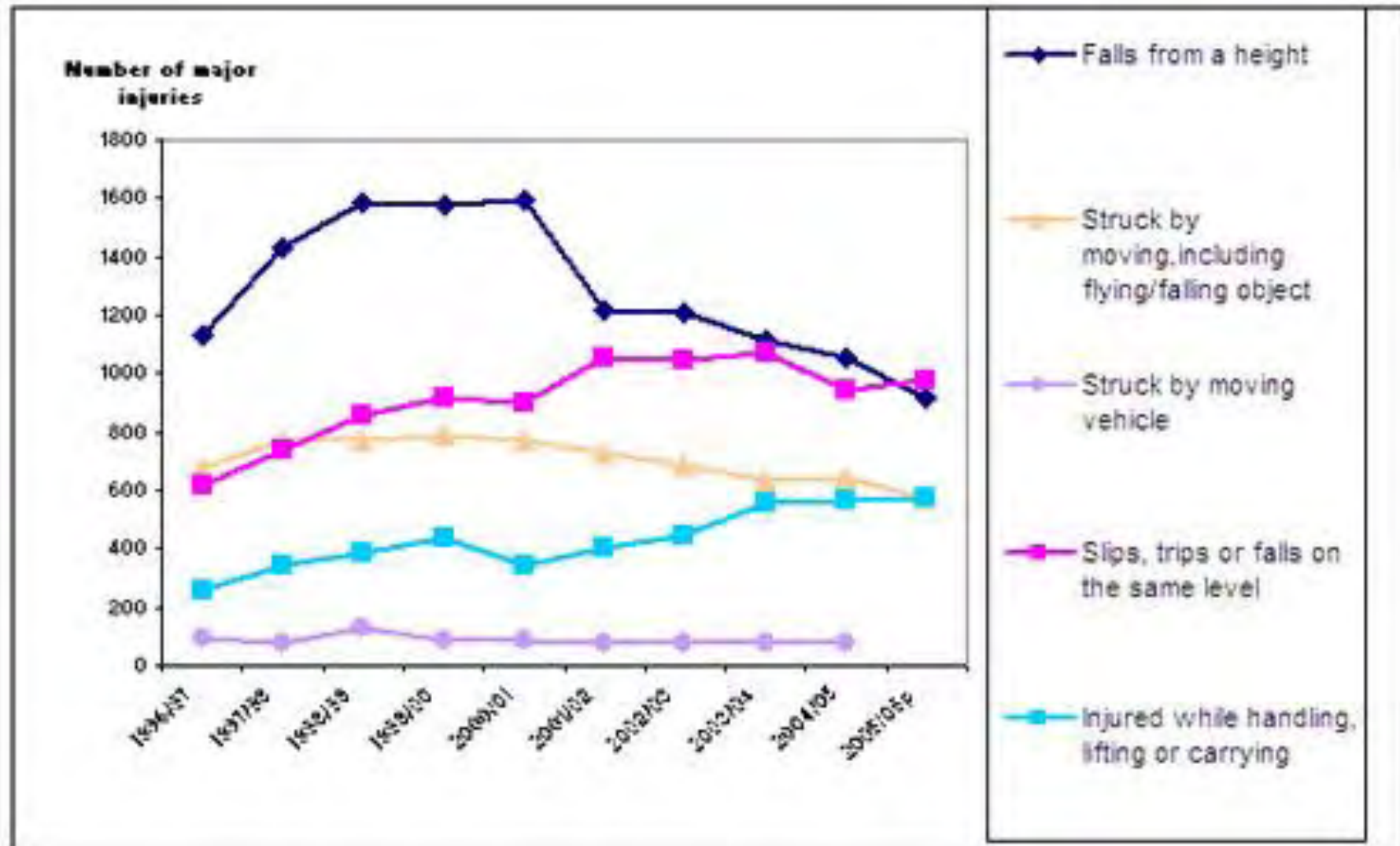
SAFETY CONSIDERATIONS DURING CONSTRUCTION STAGE



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CAUSES OF INCIDENTS



Source : OSHA publication

INCIDENT #1



INCIDENT #2



SAFETY CONSIDERATIONS IN OPERATIONS



"Here's your new work station.
Make sure you count your fingers and toes before you clock out."



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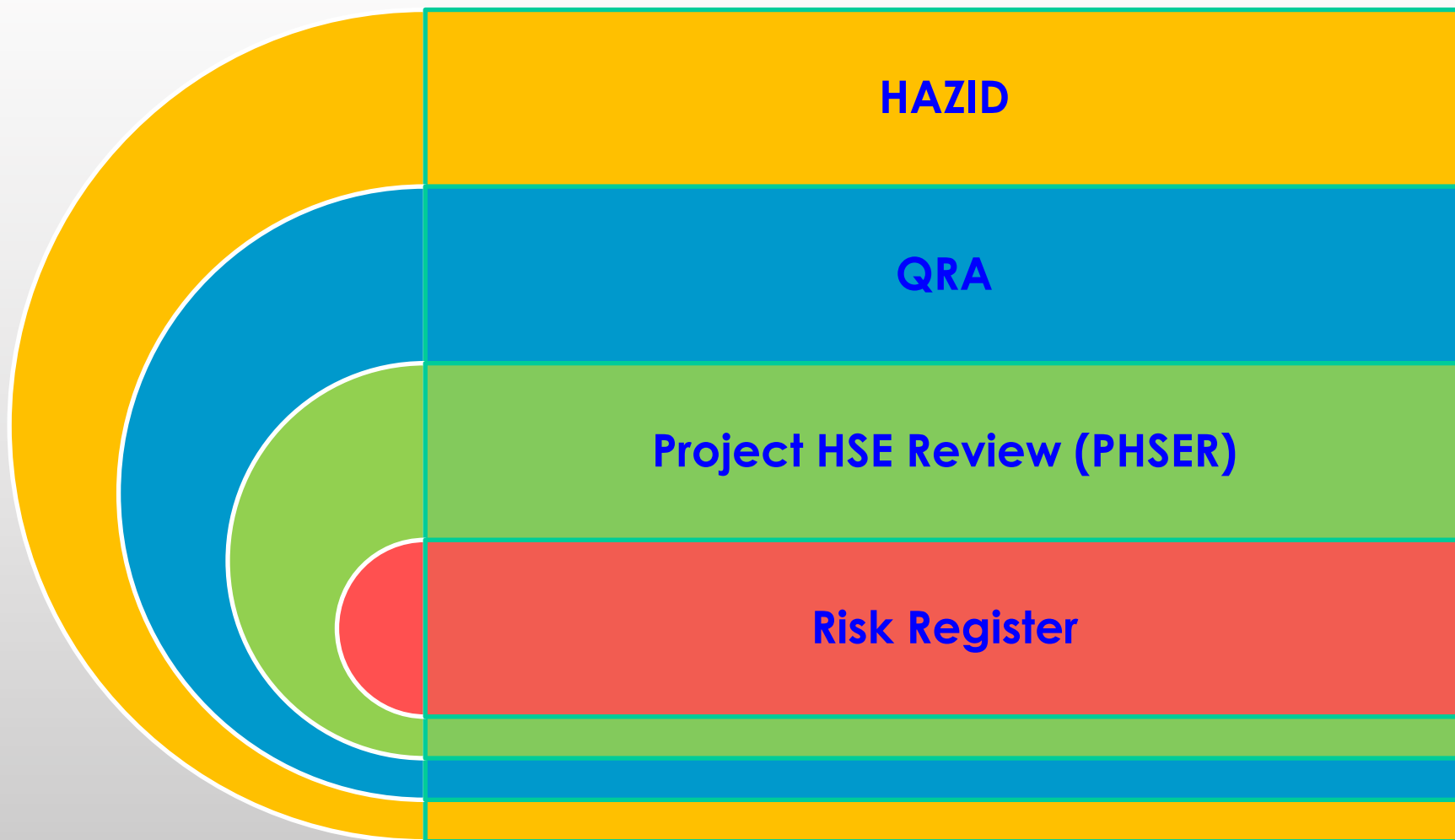
SAFETY AT OPERATIONAL STAGES



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SAFETY AT OPERATIONAL STAGES



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Safe operation with no incidents:

- Risk Based Inspection (RBI)
- Reliability Centred Maintenance (RCM)
 - Subcontractor Safety
 - Transport Safety
 - Risk Register
 - SCE/PS update
- Operations Safety Case
- Safety / PSM / SMS Audits
 - BBS



MONITORING OF OPERATIONAL PARAMETERS

- Logging of parameters in field and Panel log sheets
- Review by shift in charge / plant in charge
- Effective Process monitoring
- Availability of Real Time database management systems
- Availability of Communication Systems
 - *fPublic Address (PA) system*
 - *fWalkie Talkie*
 - *fField Telephones / Check and change room telephone*
- Close monitoring of critical equipments
- Effective use of checklists and permits
- Standard Operating Procedures



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MONITORING OF FURNACE OPERATION

- Tube skin temp., Box temp., flue gas temp.
- Any hot spots on the tube.
- Tube support / hangers condition.
- Refractory condition.
- Flame impingement.
- Vibrations, if any
- Bowing / bulging etc



EFFICIENT BURNER MANAGEMENT

SMOKEY BURNERS LEAD TO :

- POOR VISIBILITY OF FURNACE CONDITION
- LOCALIZED FLAME IMPINGEMENT
- NON-UNIFORM TEMPERATURE PROFILE
- REFRACTORY DAMAGE
- POOR EFFICIENCY



UNRELIABLE FURNACE OPERATION



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FURNACES

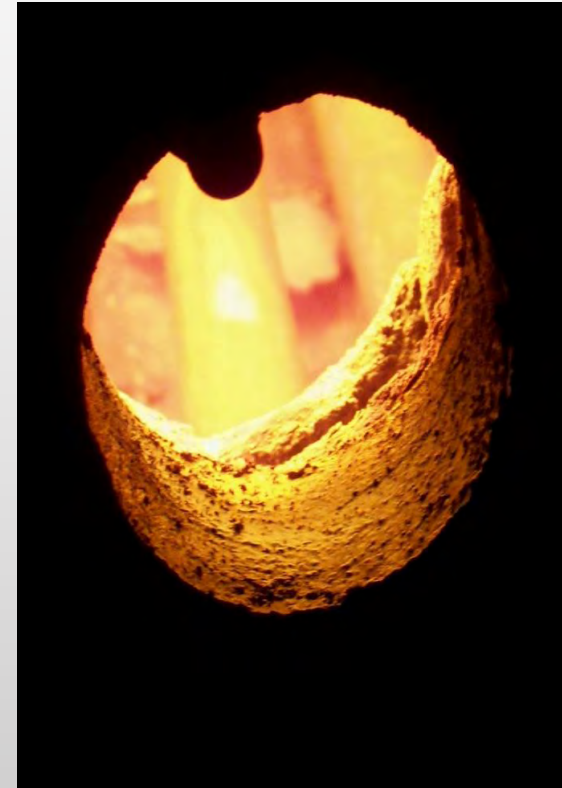
AVOID FLAME IMPINGEMENT

FLAME IMPINGEMENT MAY LEAD TO

- LOCALIZED OVERHEATING
- COKE FORMATION
- HOT SPOTS



FURNACE TUBE FAILURE



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COLUMN OVERHEAD SYSTEM MONITORING

- Column top temperature
 - Overhead pressure.
- Fluid velocity at overhead
- Dosing rate of neutralizer and corrosion inhibitor
 - Strength of solution for chemical dosing
- pH, Chloride & Fe in overhead reflux drum condensate .
 - Salt content ex Desalters
 - Wash water injection rate and quality



COLUMN OVERHEAD CORROSION

- When water condenses in column top or overhead HCl also condenses, forming strong acid. First drop conditions are the most severe.
- Neutralizer (amine or NH_4) is added to control pH to near neutral. Neutralizer salts can be corrosive.
- Recirculating water wash prevents salt accumulation and minimizes first drop effect.
- Filming amines (Corrosion inhibitor) often added to protect metal.



REACTOR HEATING/ COOLING RATES

- When a reactor is in service, hydrogen diffuses through wall.
- As reactor cools down, the solubility of hydrogen in Cr-Mo wall decreases.
- If reactor is cooled very fast, the excess hydrogen can cause cracking and cladding disbonding.
- The failure can be avoided by proper material selection and strict adherence to prescribed cooling rates.



HEAT EXCHANGERS

- Before placing any exchanger in operation, drawing and specification sheets must be referred for any special instruction.
 - **Start-up operation:** The exchangers may be placed in service by first establishing circulation of the cold fluid followed by gradual introduction of hot medium. Proper venting of all trapped air needs to be ensured.
 - **Shut-down operation:** First the flow of hot fluid is to be stopped and then that of cold fluid.
 - For fixed tube sheets, differential pressure must be maintained during start-up, shut-down and operation.
 - Thermal shock must be prevented.
 - All bolt tightening is to be ensured.



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HEAT EXCHANGERS

ON-LINE CHECKING OF HEAT EXCHANGERS

- Operating temperature & pressure.
- Tube fouling.
- Paint condition.
- Flange joint leakage.
- Condition of weld joints.
- Support conditions.



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HEAT EXCHANGERS

MAINTENANCE OF HEAT EXCHANGERS

- At regular intervals, an examination should be made of the interior and exterior condition of the unit. Sacrificial anodes should be inspected for cleaning / replacement.
- **Indication of fouling:** Exchanger subjected to fouling or scaling should be cleaned periodically.
- **Access to tubes:** To inspect inside of tubes, the channel, shell cover and floating head cover must be removed.
- **Location of leaks in tubes:** Shell test is carried out by pressurizing shell side hydraulically and identifying leaking tubes and roll leaks. Subsequent testing done for checking gasket integrity.



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STORAGE TANKS

ON-STREAM CHECKS FOR STORAGE TANKS

- Condition of painting
- Roof Plates
- Ladders, Stairways, Platforms and Structures
- Tank Pad
- Anchor Bolts
- Fire Fighting System
- Vents & Pressure Relieving Devices
- Insulation
- Grounding/ Earthing Connections
- Leaks
- Tank Mountings



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ON-STREAM CHECKS FOR PIPING

- Visual inspection for corrosion, vibrations, supports etc
- Use of NDTs for health assessment
- Corrosion measurement by probes, coupons etc.
- Inspection of underground piping
- Monitoring of functioning of CP system
- Hydrostatic testing



SAFETY AT DE-COMMISSIONING STAGES

Safe abandonment and disposal:

- HAZID
- QRA
- Project HSE Review (PHSER)
- Risk Register



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HAZARD CONTROL- DESIGN SAFETY MEASURES

QUANTITATIVE RISK ASSESSMENT (QRA)

HAZARD IDENTIFICATION METHODS:

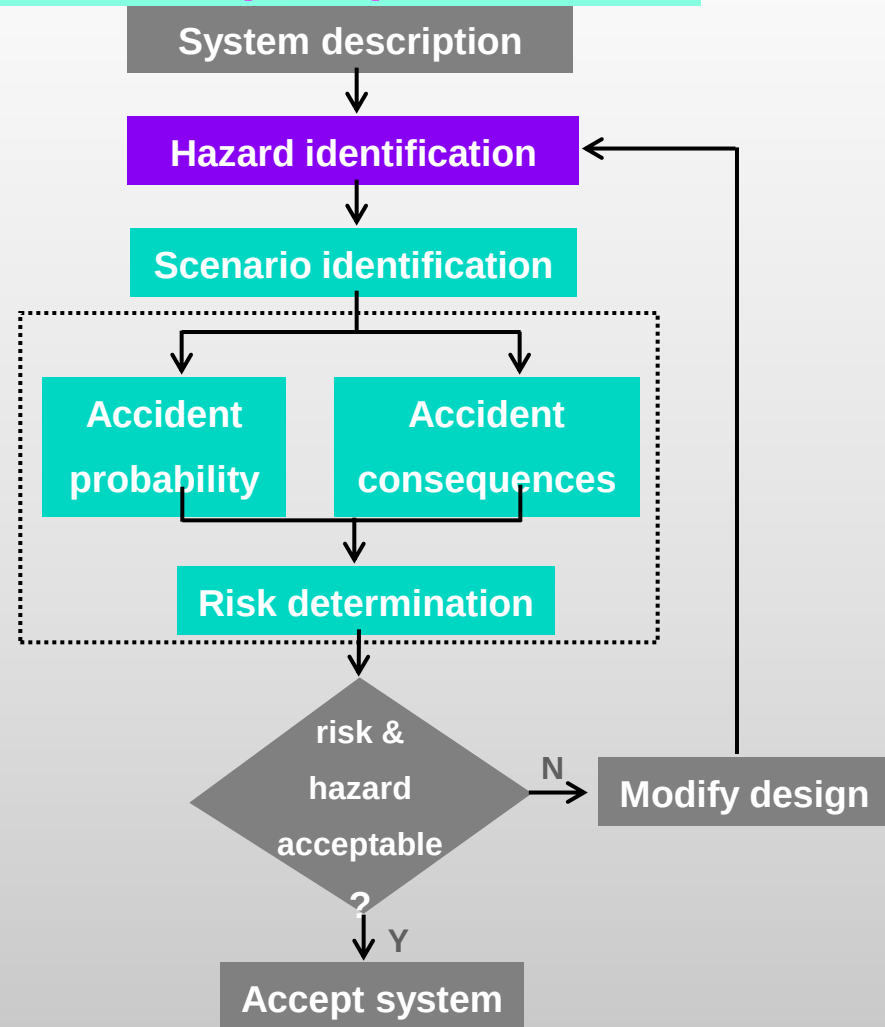
- Process hazard checklist
- Hazard survey: DOW index
- HAZOP hazard & operability study
- Safety review

RISK ASSESSMENT:

- What can go wrong & how ?
- What are the chances ?
- Consequences ?

EXTREMES

- Low probability
- Minimal consequences



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HAZARD CONTROL- DESIGN SAFETY MEASURES

BEHAVIOUR BASED SAFETY (BBS)

- BBS is a process approach to improving safety performance by helping workgroup to,
 - Identify safety-related behaviours that are critical to performance
 - Gather data on workgroup safety excellence
 - Provide ongoing, two-way performance feedback
 - Remove safety barriers to continuous improvement
- Finding out what people are doing that leads to incidents and stopping them doing it. Or...
- Finding out what people are doing to avoid incidents and getting everyone to do it
- Behaviour (unlike attitude) is visible, measurable and can be directly influenced

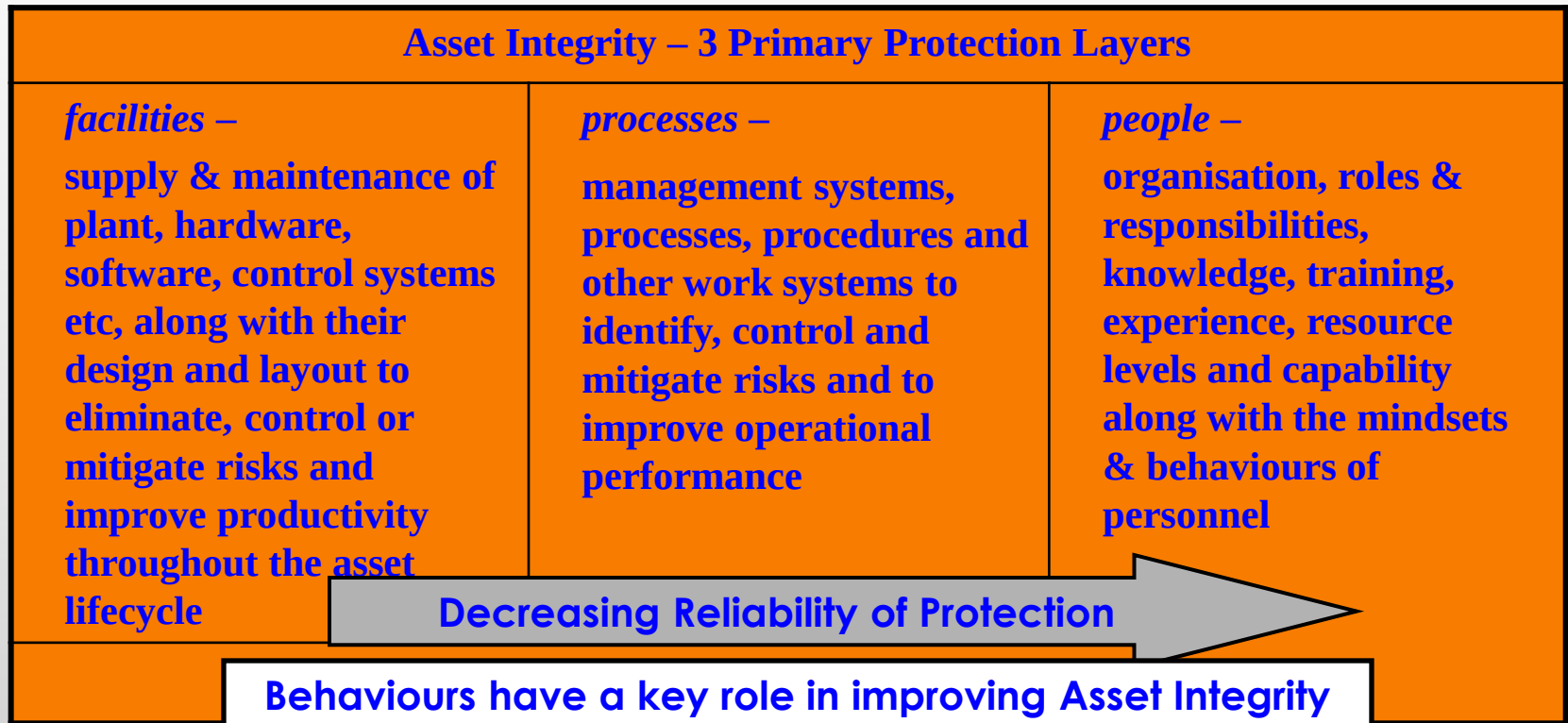


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FOOD FOR THOUGHT

To protect against major accidents and consequential losses, there are generally considered to be three layers (or types) of protection:



Both Personal Safety and Asset Integrity Safety issues can be addressed through BBS



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Thank You



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SAFETY AT PLANNING STAGE

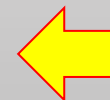
PROJECT HSE REVIEW (PHSER)

The overall objective of the PHSER (Project HSE Review) process is to assure the Company that HSE-sensitive areas have been identified in a systematic way and that the Major projects, engineering and operational systems have been or will be developed to control / manage the identified risks.

Projects from an HSE point of view are reviewed at Detail Engineering & Design, Construction , Pre-Startup and Operation Stage .



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SAFETY AT PLANNING STAGE

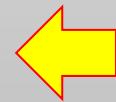
HAZARD IDENTIFICATION (HAZID)

The objectives of the HAZID procedure are to identify main hazards, to review the effectiveness of selected safety measures and, where required, to expand the safety measures in order to achieve a tolerable residual risk.

Projects from an HSE point of view are reviewed at Detail Engineering & Design, Construction , Pre-Startup and Operation Stage .



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SAFETY AT PLANNING STAGE

RISK REGISTER

A risk register is a risk management tool commonly used in risk management and compliance . It acts as a central repository for all risks identified by the organisation and, for each risk, includes information such as source, nature, treatment option, existing counter-measures, recommended counter-measures and so on. It can sometimes be referred to as a risk log.



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SAFETY AT DESIGN STAGE

HAZARD & OPERABILITY STUDY (HAZOP)

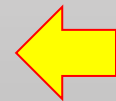
The purpose of the HAZOP is to investigate how the system or plant deviate from the design intent and create risk for personnel and equipment and operability problems.

HAZOP is used as part of a Quantitative Risk Assessment (QRA) or as a standalone analysis.

HAZOP is a more detailed review technique than HAZID.



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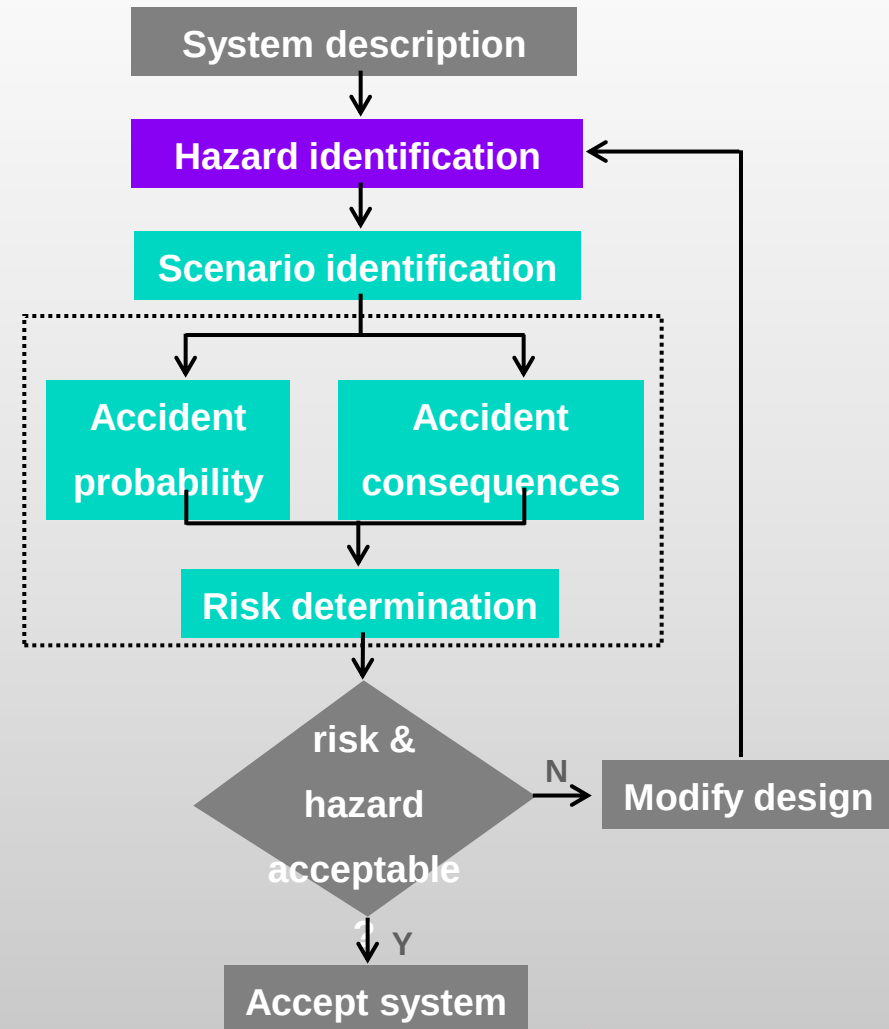
SAFETY AT DESIGN STAGE

QUANTITATIVE RISK ASSESSMENT (QRA)

QRAs provide numerical estimates of risk. These assist in understanding the risk profile of the system being analysed.

Potential improvements can then be easily identified and ranked for risk-reduction effectiveness.

The types of risk that may be considered include safety risk, environmental risk and business / operational risk. QRA is an effective tool for ranking the hazards and comparing the risk profile for various design or operational options.



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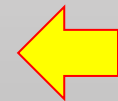
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SAFETY AT DESIGN STAGE

SAFETY INTEGRITY LEVEL (SIL)

The purpose of SIL is to take process to a “safe state” when pre-determined set points have been exceeded or when safe operating conditions have been transgressed. A SIS is comprised of safety functions with sensors, logic solvers and actuators.

A Safety Instrumented Function (SIF) is a safety function with a specified Safety Integrity Level which is implemented by a SIS in order to achieve or maintain a safe state. A SIF's sensors, logic solver, and final elements act in concert to detect a hazard and bring the process to a safe state.



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SAFETY AT DESIGN STAGE

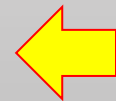
ESCAPE, EVACUATION & RESCUE ANALYSIS (EERA)

The general method for an evacuation, escape and rescue (EER) assessment are chalked out at design stage.

Typical evacuation, escape and rescue safety goals are also outlined.

The methodology used to assess the effects of fires and smoke on these EER provisions is defined.

e.g. two independent escape routes should normally be available from all areas of the platform. (This can include quantitative assessments for, e.g., smoke fire and explosion.)



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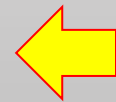


SAFETY AT DESIGN STAGE

EMERGENCY SYSTEM SURVIVABILITY ANALYSIS (ESSA)

Essential Systems Survivability Analysis (ESSA) is used to

- Identify safety critical systems
- Define the functional requirements and HSE critical element goals for each system
- Identify the Major Accident Hazard (MAH) events which could potentially stop the emergency systems functioning as required
- Evaluate each system with respect to its survivability from the Major Accident Hazards (MAHs) (which may involve some quantitative analysis).



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SAFETY AT DESIGN STAGE

SAFETY CASE

A safety case is a document produced by the operator of a facility which:

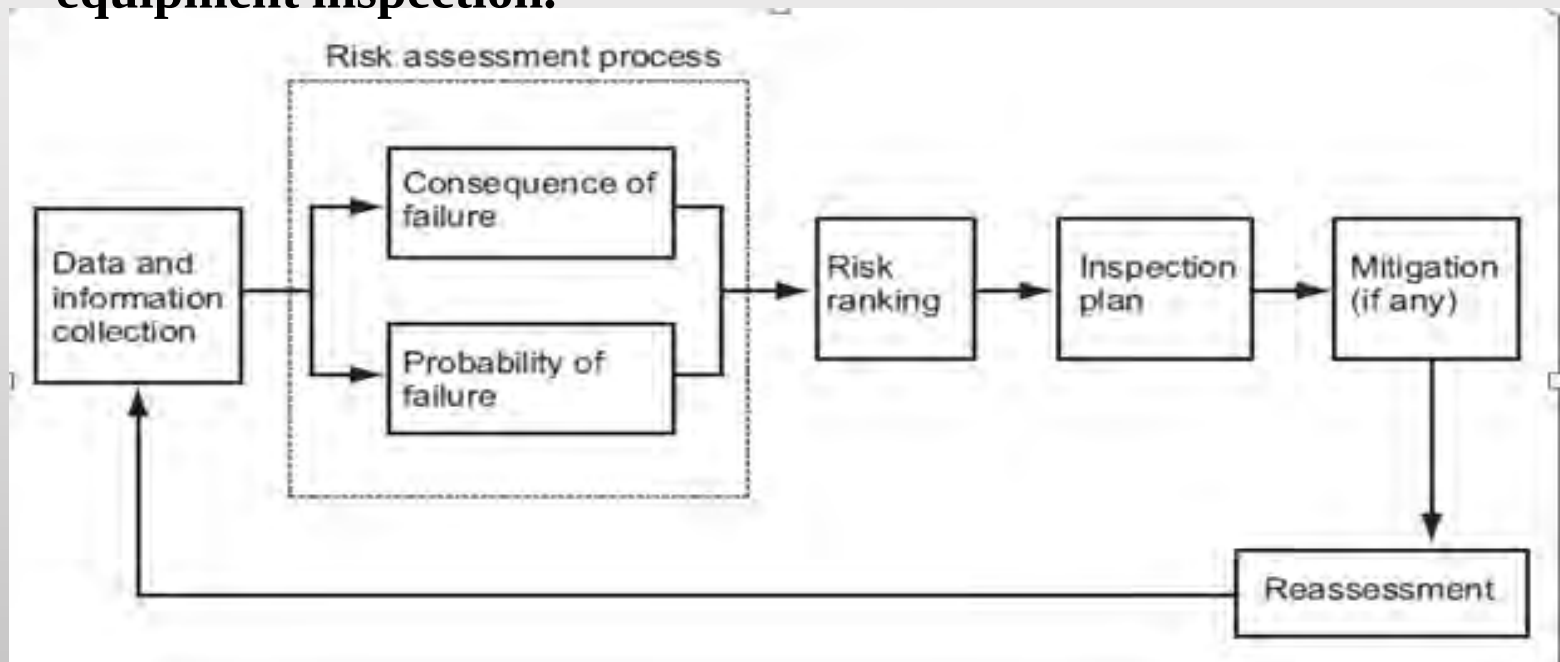
- Identifies the hazards and risks
- Describes how the risks are controlled
- Describes the safety management system in place to ensure the controls are effectively and consistently applied.



SAFETY AT OPERATION STAGE

RISK BASED INSPECTION (RBI)

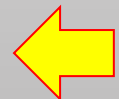
RBI is a risk assessment and management process that is focused on loss of containment of pressurized equipments in processing facilities due to material deterioration. These risks are primarily managed through equipment inspection.



RBI PLANNING PROCESS

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SAFETY AT OPERATION STAGE

RELIABILITY CENTERED MAINTENNACE (RCM)

RCM is a structured process consisting of multifunctional team of operation, maintennace and specialist with cross-functional information sharing leading to deeper understanding of equipment.

Outcomes of RCM are:

- Revised maintenance schedule and practises.
- Revised operational procedures.
- Recommendation of engineering changes
- Database of maintenance requirements.



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SAFETY AT OPERATION STAGE

OPERATIONAL SAFETY CASES

A Safety Case is a structured argument, supported by evidence, intended to justify that a system is acceptably safe for a specific application in a specific operating environment.

The review of safety cases is an important activity in the safety engineering process, performed throughout development, operation and maintenance, in the which the safety case argument and evidence are scrutinized and challenged.



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SAFETY AT OPERATION STAGE

SAFETY/ PROCESS SAFETY MANAGEMENT (PSM) / SAFETY MANAGEMENT SERVICES (SMS) AUDITS

Such audits are an analytical tool focused on cross-checking the relevance of your safety policies and their implementation.



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